

MuniGPT

Product Requirements Document -- v0.2
Instituto Iguualdad -- Santiago, Chile -- 2026

Abstract

MuniGPT is a fully local, offline-capable AI assistant designed for deployment on individual workstations within Chilean municipal institutions. It replaces the use of cloud-based AI services (ChatGPT, Gemini, Copilot) with a self-hosted large language model augmented by Retrieval-Augmented Generation (RAG) over a curated corpus of Chilean municipal and public law. The product is built on a minimal custom stack -- FastAPI backend, React frontend, Electron shell -- with no dependency on third-party AI platforms. The corpus is fetched automatically from BCN (leychile.cl) and extended at install time by IT personnel with municipality-specific documents. The product addresses cybersecurity compliance obligations arising from Ley 21.663 and ANCI/CSIRT Chile guidelines.

1. Background and Motivation

Chilean municipalities are subject to cybersecurity obligations under Ley 21.663 (2024) and its implementing regulations issued by ANCI. The use of external, cloud-hosted AI tools by public servants -- whereby institutional queries, documents, and citizen data are transmitted to foreign servers -- is inconsistent with the data sovereignty and minimal exposure principles established therein.

In practice, municipal staff have broadly adopted tools such as ChatGPT and Google Gemini for drafting correspondence, summarising legal provisions, and answering procedural questions. Prohibiting this use without offering a functional equivalent creates a compliance gap that degrades productivity and incentivises workarounds. MuniGPT addresses this gap directly.

2. Objectives

Primary. Replace cloud AI usage on municipal workstations with a fully local compliant alternative requiring no internet connection for core operation.

Secondary. Provide passive knowledge augmentation through RAG over the relevant Chilean legal corpus, automatically fetched from BCN and extended with municipality-specific documents loaded by IT personnel at setup.

Tertiary. Enable optional controlled web search, with the query string being the only data transmitted externally, logged locally for auditability.

3. Scope

Version 1.0 targets Chilean municipalities. The product is distributed as a self-contained Windows installer. Each installation is licensed per workstation and activated offline via HMAC-based license keys issued by Instituto Iguualdad. The initial target cohort is the forty lowest-scoring municipalities in the Lupa Municipal 2026 security audit.

3.1 In Scope

- Local LLM inference via Ollama (no cloud dependency for chat)
- RAG pipeline over a base legal corpus fetched from BCN at install time
- IT-loaded document extension: PDFs dropped into corpus/municipio/ at setup

- Optional web search via Brave Search API (query string only leaves the machine)
- Simple React chat UI accessible to non-technical staff
- FastAPI backend: /chat, /search, /ingest, /status, /config endpoints
- Electron shell: splash screen, process management, desktop shortcut
- Per-municipality config: name, logo
- Offline HMAC license activation
- Windows 10/11 x64 installer (Inno Setup)

3.2 Out of Scope

- Mobile or browser-based deployment
- Multi-user server deployment
- Document generation or e-signature workflows
- Integration with SIGFE, Mercado Publico, or other government platforms
- Fine-tuning or training of the base model
- Automatic corpus updates over the internet

4. System Architecture

MuniGPT is built on a minimal custom stack. There is no dependency on third-party AI platform software such as AnythingLLM, Open WebUI, or LlamaIndex. This decision minimises installer size, eliminates licence compatibility risk, and provides full control over the user experience.

4.1 Component Stack

Layer	Component	Role
Shell	Electron	Splash screen, child process management, desktop shortcut
Frontend	React + Vite (TypeScript)	Chat UI, web search toggle, citation display, municipality branding
Backend	FastAPI + Uvicorn (Python)	5 endpoints: /chat, /search, /ingest, /status, /config
Inference	Ollama	Local model serving and embedding generation
Model	Qwen 2.5 7B Instruct Q4_K_M	Best local Spanish performance; 4B fallback on low-RAM machines
Embeddings	nomic-embed-text via Ollama	Local embedding generation for RAG, no external API
Vector DB	LanceDB (embedded Python)	Hybrid BM-25 + vector search over legal corpus chunks
Corpus	corpus_fetcher.py (httpx)	Downloads Tier 0+1+2 from BCN; IT loads Tier 3 manually
Web search	Brave Search API (optional)	Only query string transmitted; result logged locally
Licensing	HMAC validator (Rust)	Offline per-workstation activation via hardware fingerprint

Table 1. MuniGPT component stack.

4.2 Data Flow

A user query enters the React chat UI. The frontend sends POST /chat to the FastAPI backend. The backend embeds the query locally via nomic-embed-text, performs hybrid BM-25 + vector search against LanceDB, and retrieves the top-5 most relevant chunks. These are prepended to the query as context and submitted to Qwen via Ollama. The response streams token-by-token via SSE back to the frontend. No data leaves the machine unless the user activates web search, in which case only the query string is transmitted to Brave Search over HTTPS.

4.3 Hardware Requirements

Tier	RAM	GPU	Model	Speed
Minimum	8 GB	None (CPU)	Qwen 2.5 4B Q4	3-6 tok/s
Recommended	16 GB	None/iGPU	Qwen 2.5 7B Q4	5-12 tok/s
Optimal	16 GB	dGPU 6+ GB	Qwen 2.5 7B Q4	20-40 tok/s

Table 2. Hardware tiers. OptiPlex-class machines typically fall in Minimum-Recommended.

5. Legal Corpus

The RAG knowledge base is structured in four tiers. Tiers 0-2 are downloaded automatically from BCN by corpus_fetcher.py. Tier 3 is loaded manually by IT personnel during on-site setup. All tiers are ingested into LanceDB by ingest.py. The resulting database ships pre-built inside the installer to eliminate any ingestion delay on first launch.

Tier 0 -- General Public Law (7 documents)

Applies to all public bodies. Includes the Constitucion Politica (Cap. XIV municipios), Ley 18.575 Bases Administracion del Estado, Ley 19.880 Procedimientos Administrativos, Ley 20.285 Transparencia, Ley 20.500 Participacion Ciudadana, Ley 21.663 Ciberseguridad, and Ley 21.180 Transformacion Digital del Estado.

Tier 1 -- Core Municipal Law (10 documents)

Daily operational law for municipal staff. Includes DFL 1/2006 LOC Municipalidades, Ley 18.883 Estatuto Funcionarios Municipales, DL 3.063 Rentas Municipales, Ley 19.418 Juntas de Vecinos, Ley 15.231 Juzgados de Policia Local, Ley 19.886 Compras Publicas, DS 250 Reglamento Compras, DS 458 LGUC, Ley 19.378 Atencion Primaria Salud, and Ley 20.730 Lobby.

Tier 2 -- Extended (7 documents)

Complementary law for municipalities with specific profiles. Includes DFL 1/1997 Estatuto Docentes, Ley 21.040 Nueva Educacion Publica, Ley 21.591 Royalty Minero, DS 854 Clasificaciones Presupuesto, Ley 19.925 Expendio Alcohol, Ley 20.965 Consejo Seguridad Comunal, and DFL 1/2003Codigo del Trabajo.

Tier 3 -- IT-Loaded Documents (variable)

During on-site setup, IT personnel drop additional PDFs into corpus/municipio/ before running ingest.py. These may include the Ordenanza Municipal, Reglamento Interno, PLADECO, organigrama, circulares SUBDERE relevantes, and any other institutional documents. Documents in this tier are labelled with their source filename in every RAG citation returned to the user. Future updates -- for example, when an ordenanza is amended -- are handled by replacing the PDF and re-running ingest.py without reinstalling the application.

6. Functional Requirements

ID	Requirement	Priority
FR-01	The system shall operate entirely locally without requiring internet access for chat and RAG functionality.	Must
FR-02	The system shall answer queries in Spanish as the default language.	Must
FR-03	The system shall cite the source document filename and chunk when a response is grounded in the RAG corpus.	Must
FR-04	Chat responses shall be streamed token-by-token via SSE to maintain perceived responsiveness on low-spec hardware.	Must
FR-05	The system shall allow the user to toggle web search on or off per session via a visible UI control.	Should
FR-06	Web search shall transmit only the query string externally. No document content, chat history, or personal data shall leave the machine.	Must
FR-07	All web search queries shall be logged locally with timestamp, query text, and result count.	Should
FR-08	The system shall activate via an offline HMAC license key tied to the machine hardware fingerprint.	Must
FR-09	corpus_fetcher.py shall download all Tier 0, 1, and 2 documents from BCN during installation without manual intervention.	Must
FR-10	IT personnel shall be able to load additional PDFs into corpus/municipio/ at setup time without modifying any application code or configuration.	Must
FR-11	ingest.py shall process all PDFs across all corpus tiers in a single invocation and produce a LanceDB database ready for immediate use.	Must
FR-12	IT-loaded documents (Tier 3) shall be labelled with their source filename in RAG citations displayed to the user.	Must
FR-13	Re-running ingest.py after adding new documents to corpus/municipio/ shall update the corpus without reinstalling the application.	Must
FR-14	The installer shall bundle all components in a single .exe with no prerequisites required on the target machine.	Must
FR-15	The system shall auto-select the 4B model if available RAM is below 12 GB at launch.	Could

Table 3. Functional requirements (MoSCoW prioritisation).

7. Non-Functional Requirements

Performance. First token latency under 8 seconds on minimum-spec hardware. Sustained generation at no less than 3 tokens per second on CPU-only machines.

Privacy. Zero data transmission during normal operation. Web search is the sole exception: only the query string is transmitted, never document content or chat history.

Compliance. Architecture must satisfy the minimal exposure and data localisation principles of Ley 21.663 and ANCI guidelines for municipal entities.

Reliability. The application must recover automatically after system restart without requiring user reconfiguration.

Usability. A non-technical municipal employee must be able to conduct a productive session within five minutes of first launch with no training beyond a one-page quick-start card.

IT Setup. An IT installer with basic Windows knowledge must be able to complete the full setup -- corpus download, document loading, ingestion, and license activation -- within two hours on a clean machine.

Maintainability. The corpus must be updatable by dropping PDFs into corpus/municipio/ and re-running ingest.py, without reinstalling the application.

Installer size. The full bundle must not exceed 8 GB to be distributable via USB drive at installation visits.

8. IT Setup Process

The following sequence describes the complete on-site installation. Steps 1-2 are automated by the Inno Setup installer. Steps 3-6 require manual action by IT personnel.

Step	Action	Who	Est. time
1	Run MuniGPT .exe. Installs Ollama, pulls model weights, installs FastAPI backend and React frontend.	Automated	15-30 min
2	corpus_fetcher.py runs automatically, downloading Tier 0+1+2 from BCN.	Automated	5-15 min
3	IT copies municipality-specific PDFs (ordenanza, reglamento, PLADECO, circulares) into corpus/municipio/.	IT personnel	5-10 min
4	IT runs: python ingest.py. All corpus tiers are chunked, embedded, and loaded into LanceDB.	IT personnel	5-15 min
5	IT enters HMAC license key when prompted. System validates against hardware fingerprint.	IT personnel	1 min
6	IT launches MuniGPT, verifies chat responds to a test query in Spanish, hands over to staff.	IT personnel	5 min

Table 4. IT setup sequence. Total estimated time: 40-75 minutes per workstation.

9. Commercial and Licensing Model

MuniGPT is distributed by Instituto Igualdad under a non-commercial institutional model. Key open-source licences in use: Qwen 2.5 (Tongyi Qianwen Licence 2.0, commercial use permitted), Ollama (MIT), LanceDB (Apache 2.0), FastAPI (MIT), React (MIT), Electron (MIT).

Item	Amount (USD)	Notes
Donation to Instituto Igualdad	100	Per workstation licence.
Installation and configuration fee	100-150	Includes corpus download, IT document loading, ingestion, and staff walkthrough.
Annual corpus update (optional)	50	Updated base corpus + ingest re-run, delivered on-site or via USB.

Table 5. Per-installation pricing structure.

10. Delivery Milestones

Milestone	Deliverable	Target
M0 -- Stack	FastAPI + Ollama + LanceDB. Chat loop verified via curl.	Week 1

Milestone	Deliverable	Target
M1 -- Corpus + RAG	corpus_fetcher.py downloads base corpus. ingest.py loads all tiers including IT documents. RAG validated on 15 test queries.	Week 2
M2 -- Frontend	React chat UI with streaming, citations, web search toggle, branding.	Week 3
M3 -- Electron	Splash screen, process management, desktop shortcut.	Week 4
M4 -- Pilot	Installation at 2-3 pilot municipalities. IT document loading tested on-site.	Week 8
M5 -- v1.0	Inno Setup installer. HMAC licensing. Auto model tier selection. Audit logging.	Week 14

Table 6. Delivery milestones. Weeks relative to project kickoff.

11. Risks and Mitigations

Risk	Likelihood	Mitigation
BCN changes PDF export URL	Low	corpus_fetcher.py validates content-type on each download. Base corpus can also ship pre-downloaded in the installer.
IT loads low-quality or scanned PDFs	Medium	ingest.py logs every file and its chunk count. Setup guide specifies text-searchable PDF requirement. Scanned PDFs produce zero chunks and are flagged.
Hardware heterogeneity	Medium	Auto model tier selection at launch based on available RAM. Documented minimum spec. Tested on 8 GB OptiPlex-class hardware.
Model hallucination on legal citations	Medium	RAG citation display shows source filename and chunk. System prompt instructs model to answer only from provided context.
Corpus staleness	Medium	Annual corpus update service. Corpus fetch date displayed in UI. IT can update Tier 3 independently by replacing PDFs and re-running ingest.py.

Table 7. Risk register.

12. Open Questions

1. Should web search be disabled by default and require explicit IT activation, or should it be user-toggleable from the chat UI?
2. Is a central audit log aggregation mechanism required by ANCI, or is per-workstation local logging sufficient?
3. Should the installer support silent/GPO deployment for IT departments managing multiple workstations?
4. Should ingest.py support OCR for scanned PDFs that IT personnel may provide, or should the setup guide require text-searchable PDFs only?
5. What is the correct process for corpus updates when a municipality amends its ordenanza -- does IT re-run ingest.py independently or request a maintenance visit?